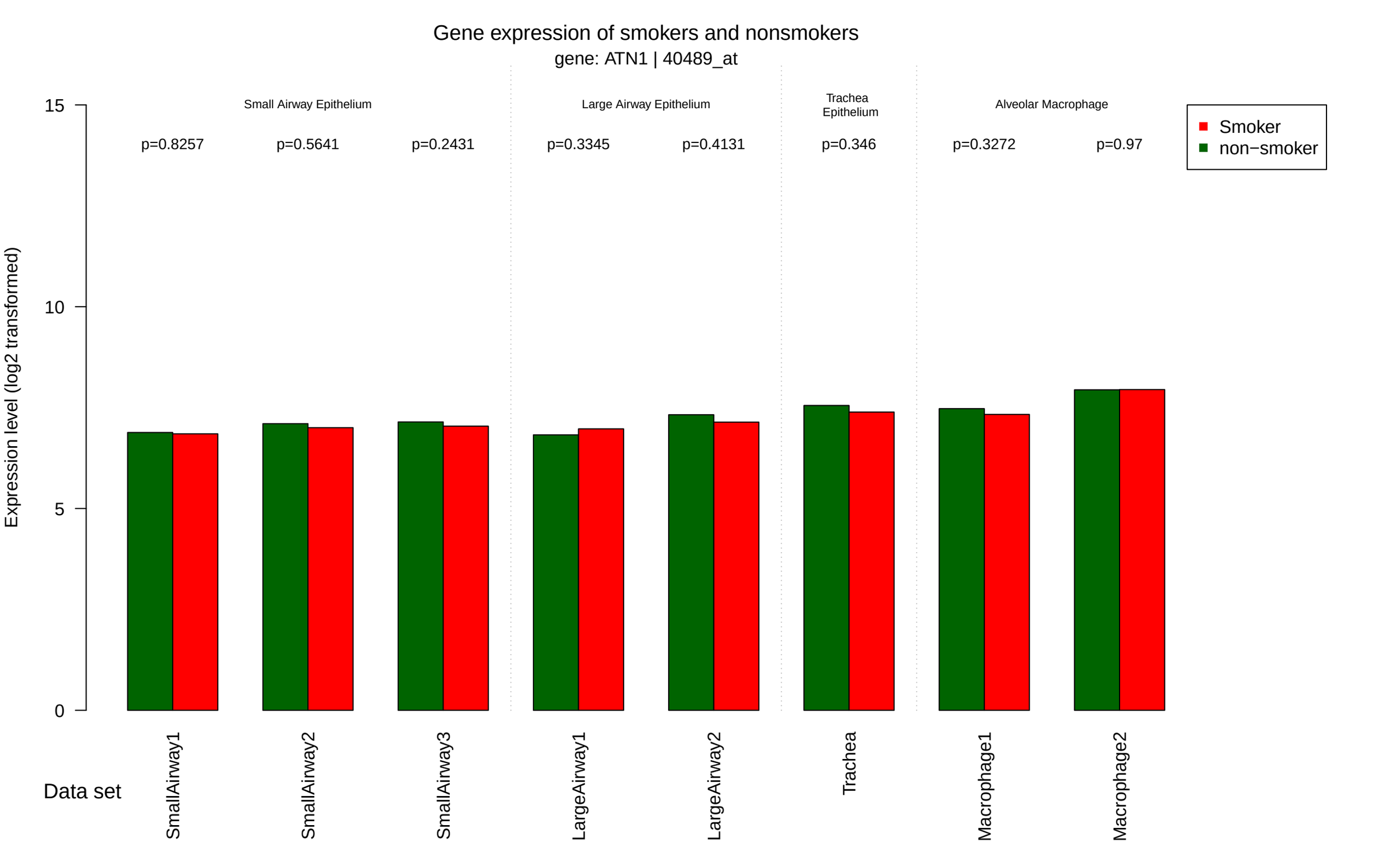
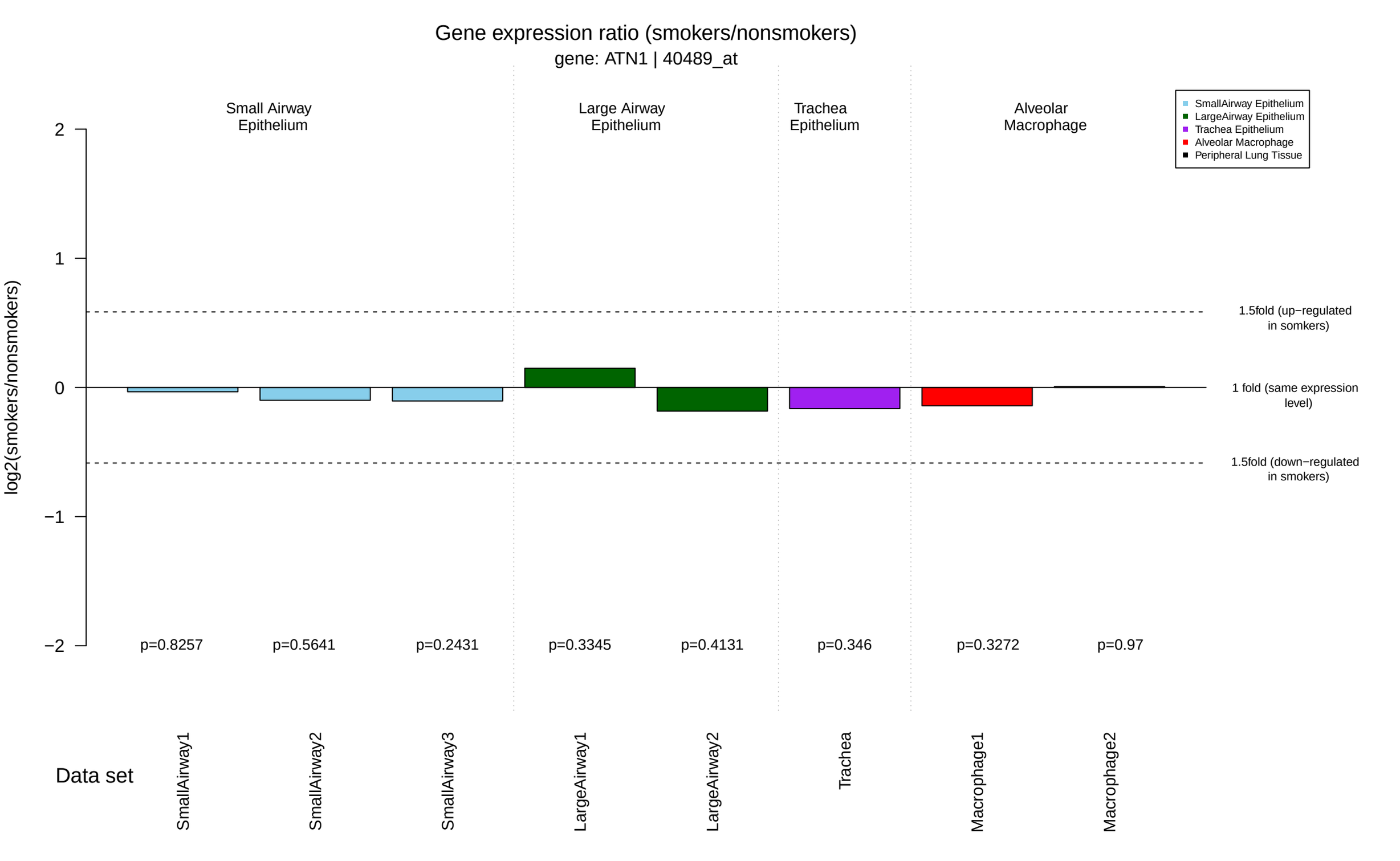
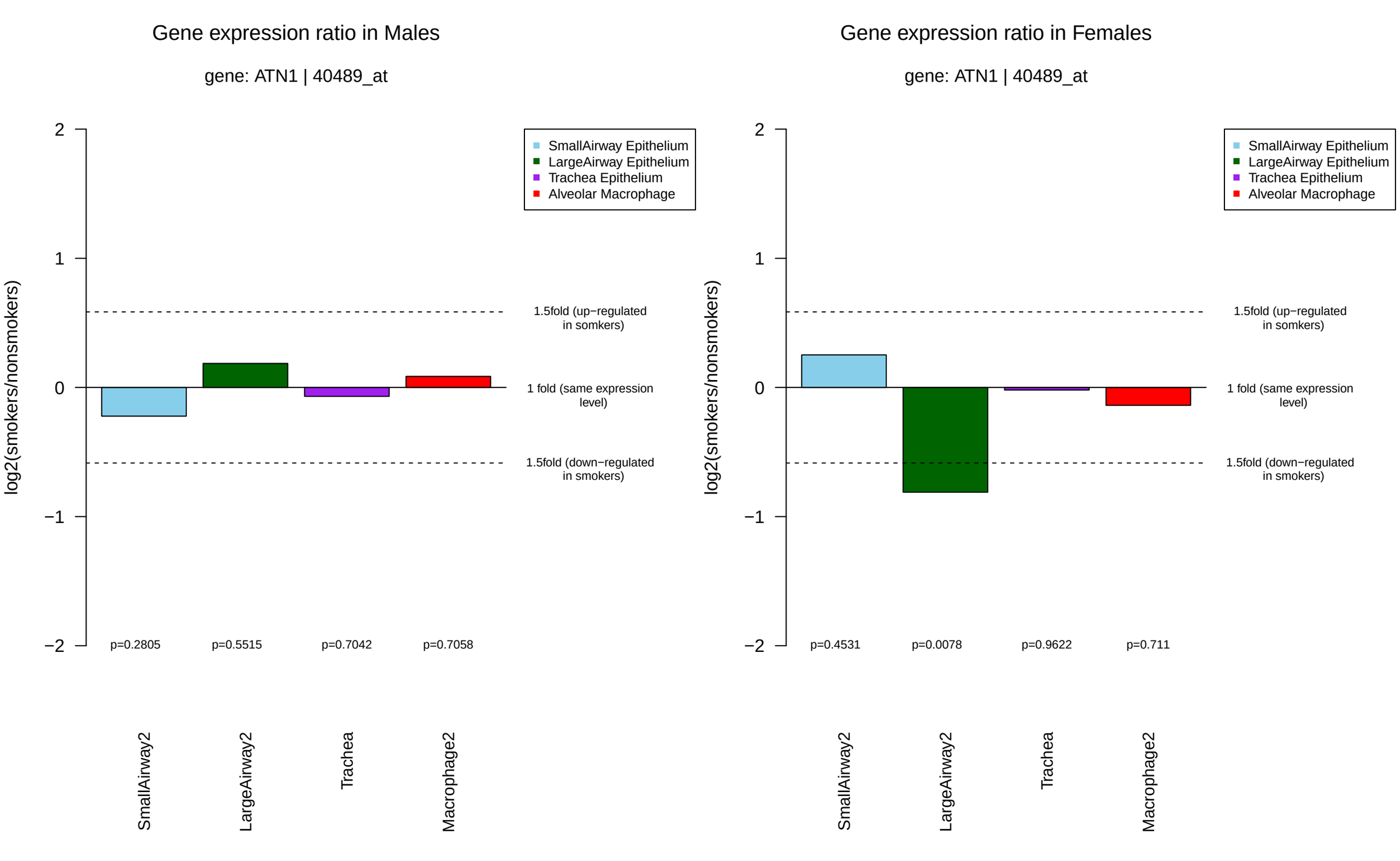
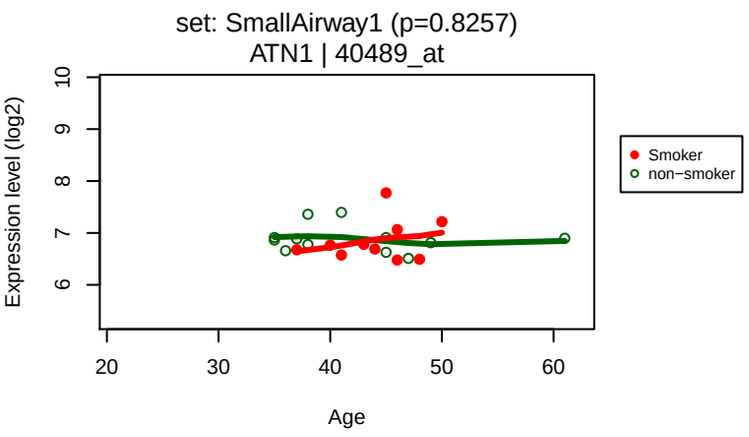
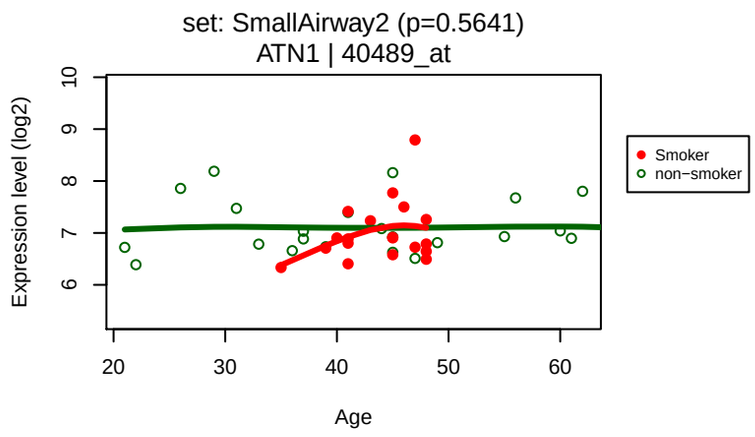
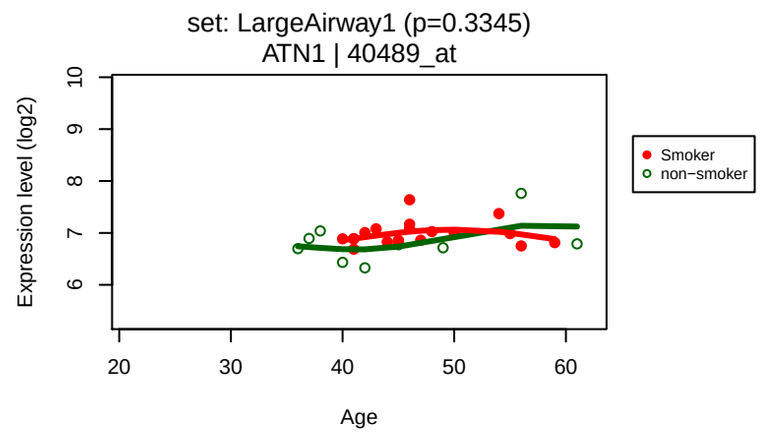

Gene expression of smokers and nonsmokers
gene: ATN1 | 40489_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8257
p=0.5641
p=0.2431
p=0.3345
p=0.4131
p=0.346
p=0.3272
p=0.97
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

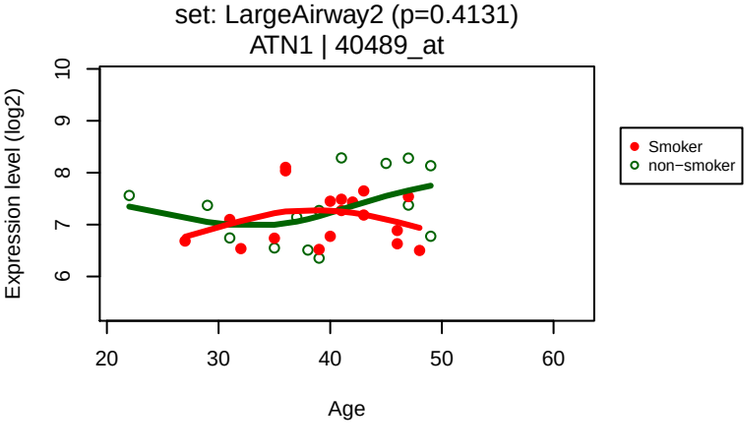

Gene expression ratio (smokers/nonsmokers)
gene: ATN1 | 40489_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8257
p=0.5641
p=0.2431
p=0.3345
p=0.4131
p=0.346
p=0.3272
p=0.97
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

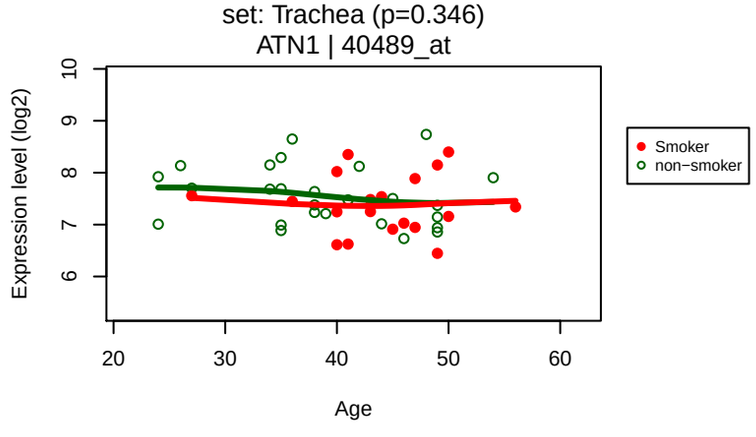

Gene expression ratio in Males
gene: ATN1 | 40489_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2805
p=0.5515
p=0.7042
p=0.7058
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATN1 | 40489_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4531
p=0.0078
p=0.9622
p=0.711
SmallAirway2
LargeAirway2
Trachea
Macrophage2

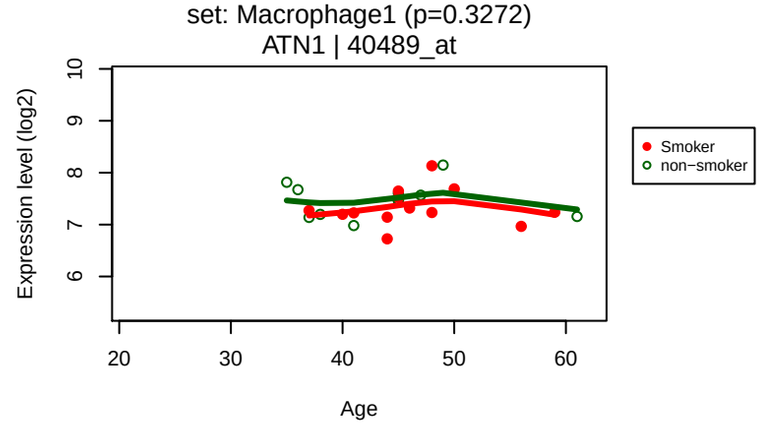

set: SmallAirway1 (p=0.8257)
ATN1 | 40489_at
Expression level (log2)
Age
Smoker
non−smoker

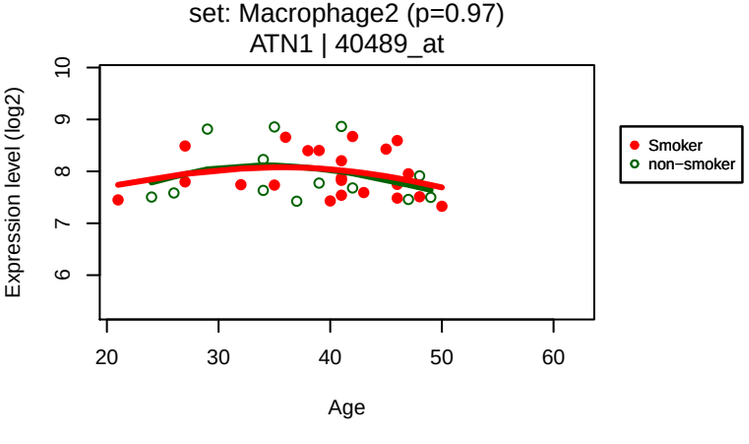

set: SmallAirway2 (p=0.5641)
ATN1 | 40489_at
Expression level (log2)
Age
Smoker
non−smoker

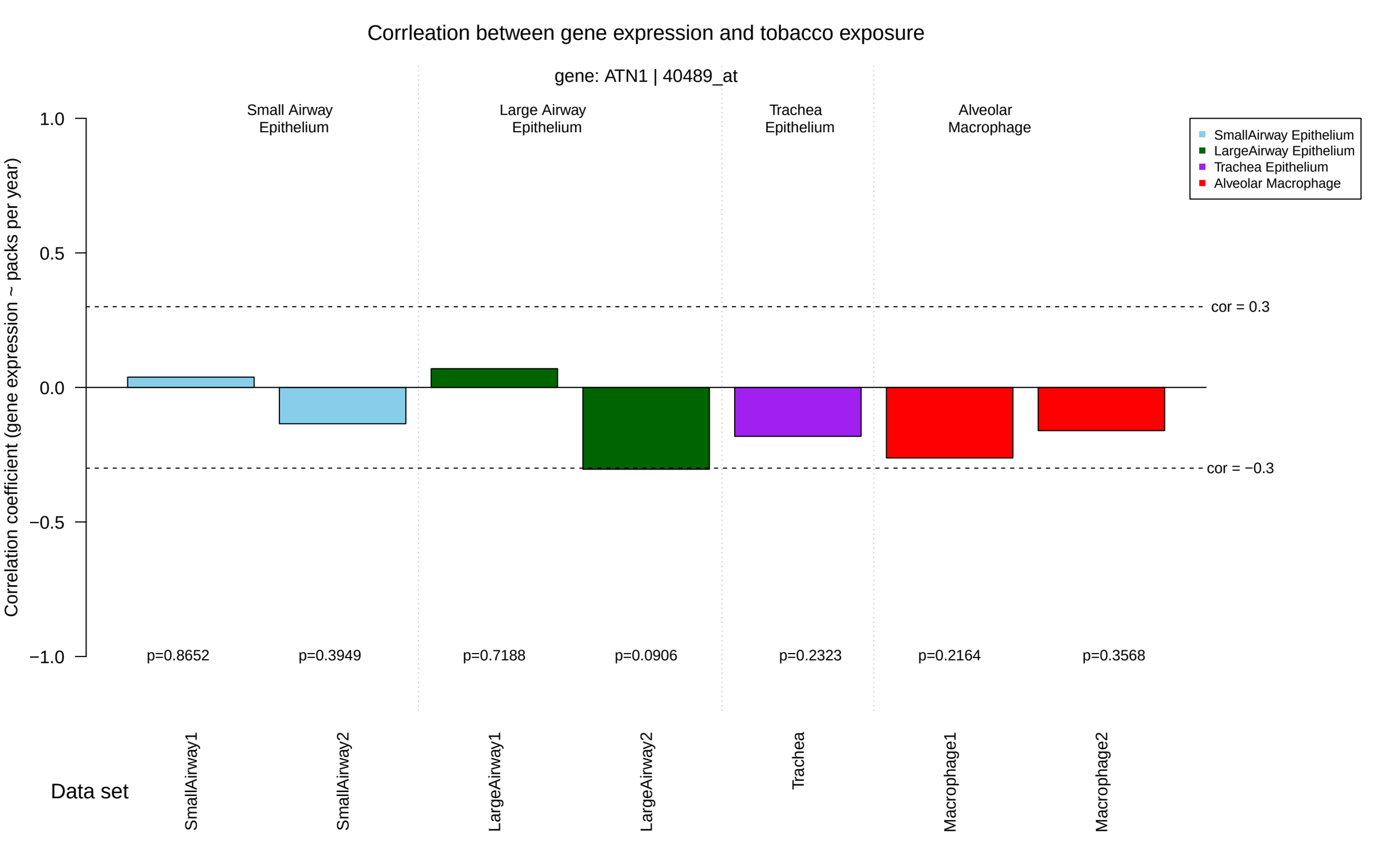

set: LargeAirway1 (p=0.3345)
ATN1 | 40489_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4131)
ATN1 | 40489_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.346)
ATN1 | 40489_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3272)
ATN1 | 40489_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.97)
ATN1 | 40489_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATN1 | 40489_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8652
p=0.3949
p=0.7188
p=0.0906
p=0.2323
p=0.2164
p=0.3568
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2